

Topology General Exam

August 16, 2010

Name:

Instructions: This is a four hour exam and 'closed book'. There are eight problems.

1. **Problem 1**

- (a) Let $T \subset \mathbb{R}^5$ be a closed subspace homeomorphic to $\mathbb{R}^2$. Explain why T will be a retract of $\mathbb{R}^5$.
- (b) View S^n as $\mathbb{R}^n \cup \{\infty\}$, so that the open subsets of S^n containing ∞ are precisely the complements of compact subsets of $\mathbb{R}^n$. Recall that a continuous function $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ is called proper if $f^{-1}(C)$ is compact in $\mathbb{R}^m$ whenever C is compact in $\mathbb{R}^n$. Show that such a proper map extends uniquely to a continuous function $\bar{f} : S^m \rightarrow S^n$.
- (c) With T as in part (a), check that the inclusion $i : T \hookrightarrow \mathbb{R}^5$ is proper. By contrast, show that no retraction $r : \mathbb{R}^5 \rightarrow T$ can be proper. (Hint: start by using part (b).)

2. **Problem 2**

Let $\mathbb{R}^\infty$ denote the union $\mathbb{R} \hookrightarrow \mathbb{R}^2 \hookrightarrow \mathbb{R}^3 \hookrightarrow \dots$, with the union topology, i.e. $U \subset \mathbb{R}^\infty$ is open iff $U \cap \mathbb{R}^n$ is open in $\mathbb{R}^n$ for all n . Let $\mathbb{R}^\omega$ denote the product of a countable number of copies of $\mathbb{R}$, with the product topology. Check that the evident set theoretic inclusion $i : \mathbb{R}^\infty \rightarrow \mathbb{R}^\omega$ is continuous, but is not a homeomorphism onto its image.

3. **Problem 3**

- (a) Describe a connected double cover of $\mathbb{R}P^2 \vee \mathbb{R}P^2$. (There is more than one correct answer.)
- (b) What are the homology groups of your double cover?
- (c) What is the fundamental group of your double cover?

4. **Problem 4**

Let M be the compact surface with boundary circle C as pictured:

- (a) Explain why M is homotopy equivalent to a figure eight. (Hint: M is the torus with a disk removed, and the torus is often represented as a square with opposite edges identified.)

- (b) Explain why the inclusion $i : C \hookrightarrow M$ induces the zero homomorphism from $H_1(C)$ to $H_1(M)$.
- (c) By contrast, explain why i is not null homotopic.

5. **Problem 5**

Suppose given a commutative diagram of abelian groups

$$\begin{array}{ccccccc}
 & & & 0 & & 0 & \\
 & & & \downarrow & & \downarrow & \\
 & & & B_1 & \xrightarrow{\beta} & B_2 & \\
 & & & \downarrow & & \downarrow & \\
 0 & \longrightarrow & A_1 & \longrightarrow & C & \longrightarrow & D \longrightarrow 0 \\
 & & \downarrow \alpha & & \downarrow & & \downarrow \\
 0 & \longrightarrow & A_2 & \longrightarrow & E & \longrightarrow & F \longrightarrow 0 \\
 & & & & \downarrow & & \downarrow \\
 & & & & 0 & & 0
 \end{array}$$

The rows are $0 \rightarrow A_1 \rightarrow C \rightarrow D \rightarrow 0$ and $0 \rightarrow A_2 \rightarrow E \rightarrow F \rightarrow 0$. The vertical columns are $0 \rightarrow B_1 \rightarrow C \rightarrow E \rightarrow 0$ and $0 \rightarrow B_2 \rightarrow D \rightarrow F \rightarrow 0$, with arrows downward. The additional maps are α from A_1 to A_2 and β from B_1 to B_2 . All rows and columns are exact, and all squares commute.

with exact rows and columns. Show that there are isomorphisms $\ker \alpha \simeq \ker \beta$ and $\operatorname{coker} \alpha \simeq \operatorname{coker} \beta$.

6. **Problem 6**

Recall that an n -dimensional manifold is a Hausdorff topological space M that can be covered by open sets homeomorphic to open sets in $\mathbb{R}^n$. Prove that a compact n -dimensional manifold can be embedded in (i.e. is homeomorphic to a subset of) $\mathbb{R}^N$ for large enough N . (Hint: use a partition of unity associated to a finite open cover $U_1, \dots, U_k$ of M equipped with embeddings $f_i : U_i \rightarrow \mathbb{R}^n$.)

7. **Problem 7**

Let $C \subset \mathbb{R}^3$ be the union of the x -axis and the y -axis. Compute $H_* (\mathbb{R}^3 - C)$. (Hint: note that $\mathbb{R}^3 - C = (\mathbb{R}^3 - x\text{-axis}) \cap (\mathbb{R}^3 - y\text{-axis})$.)

8. **Problem 8**

Let X be a Hausdorff space, and $f : X \rightarrow X$ a continuous function such that

- $f(x) \neq x$ for all $x \in X$, and
 - $f \circ f$ is the identity.
- (a) Show that every $x \in X$ has an open neighborhood W_x satisfying $f(W_x) \cap W_x = \emptyset$.
- (b) Let $\overline{X} = X / (x \sim f(x))$, with the quotient space topology. Show that the quotient map $q : X \rightarrow \overline{X}$ is a covering map.